NHL Hockey Analytics Database

SQL Server Schema & Implementation Specification — Revised Analytics Architecture

Purpose: provide Claude Code with a concrete specification for creating a local SQL Server database capable of storing NHL game data and reconstructing a comprehensive hockey analytics dataset. The system should support official NHL player statistics as well as derived individual-player, on-ice, and goalie analytics such as Corsi, Fenwick, expected goals (xG), and goals saved above expected.

1. Design Goals

• **Events are the source of truth.** Store granular play-by-play, shot, goal, shift, and on-ice-player data so statistics can be recalculated.

• **Store official statistics.** NHL-provided player game and season statistics remain first-class data.

• **Calculate advanced statistics from events.** Do not depend on a single precomputed API statistic when it can be reconstructed from the underlying events.

• **Materialize expensive analytics.** Calculate advanced statistics during ingestion/recalculation and store them for fast querying.

• **Keep calculations reproducible.** Every model-based metric must identify its calculation/model version.

• **Support three analytics families.** Individual player statistics, player on-ice statistics, and goalie statistics should be modeled separately.

• **Use NHL IDs as external identifiers.** Use internal surrogate keys for relational joins.

• **Make ingestion idempotent.** Re-running an import should update existing records rather than create duplicates.

• **Do not store video data.** This design intentionally excludes video/media tables.

2. Analytics Architecture

```
                    NHL API
                       ■
          ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
          ■                        ■
    Game Metadata            Play-by-Play
          ■                        ■
          ▼                        ▼
    Games / Teams              Plays
                                   ■
           ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
           ■                       ■                       ■
           ▼                       ▼                       ▼
         Shots                   Goals                   Shifts
           ■
           ▼
   Shot Characteristics
           ■
           ▼
      Shot-Level xG
           ■
     ■■■■■■■■■■■■■■■■■■
     ■             ■
     ▼             ▼
Individual Analytics  On-Ice Analytics
     ■             ■
     ▼             ▼
```

```
PlayerGameAdvanced  PlayerGameOnIceStats
      ■               ■
      ■■■■■■■■■■■■■■■■■
              ▼
      Goalie Analytics
              ■
              ▼
    GoalieGameAdvancedStats
```

3. Analytics Categories

Category	Examples	Primary Data Sources
Official player stats	Goals, assists, points, shots, TOI, hits, blocks, faceoffs	NHL box score/stat feeds
Individual player analytics	Individual Corsi, Fenwick, individual xG, scoring chances, high-danger chances	Shots + plays + player identity
On-ice analytics	CF/CA, CF%, FF/FA, xG For/Against, GF/GA, PDO, high-danger chances	Shots + goals + on-ice players + shifts
Goalie analytics	SA, SV, GA, SV%, xGA, GSAx, high-danger shots/saves	Shots + goalie identity + xG
Shot analytics	Distance, angle, shot type, rebound, rush, xG	Shot events + xG model

4. SQL Server Schemas

Schema	Purpose
nhl.Reference	Seasons, teams, players, player/team history, situations
nhl.Game	Games, plays, shots, goals, shifts, on-ice player state
nhl.Stats	Official NHL player game and season statistics
nhl.Analytics	Derived individual, on-ice, goalie, shot, and model-based statistics
nhl.Ingestion	Raw API responses and ingestion/reprocessing status

5. Reference Tables

Seasons

```
CREATE TABLE nhl.Reference.Seasons
(
    SeasonID        INT IDENTITY(1,1) NOT NULL,
    NHLSeasonID     INT NOT NULL,
    StartYear       SMALLINT NOT NULL,
    EndYear         SMALLINT NOT NULL,
    DisplayName     VARCHAR(9) NOT NULL,
    CONSTRAINT PK_Seasons PRIMARY KEY (SeasonID),
    CONSTRAINT UQ_Seasons_NHLSeasonID UNIQUE (NHLSeasonID)
);
```

Teams

```
CREATE TABLE nhl.Reference.Teams
(
    TeamID          INT IDENTITY(1,1) NOT NULL,
    NHLTeamID       INT NOT NULL,
    Abbreviation    VARCHAR(5) NOT NULL,
    TeamName        VARCHAR(100) NOT NULL,
    Location        VARCHAR(100) NULL,
    Conference      VARCHAR(20) NULL,
    Division        VARCHAR(30) NULL,
    Active          BIT NOT NULL DEFAULT 1,
    CONSTRAINT PK_Teams PRIMARY KEY (TeamID),
    CONSTRAINT UQ_Teams_NHLTeamID UNIQUE (NHLTeamID)
);
```

Players

```
CREATE TABLE nhl.Reference.Players
(
    PlayerID        BIGINT IDENTITY(1,1) NOT NULL,
    NHLPlayerID     INT NOT NULL,
    FirstName       VARCHAR(100) NULL,
    LastName        VARCHAR(100) NULL,
    FullName        VARCHAR(200) NOT NULL,
    PositionCode    VARCHAR(5) NULL,
    Shoots          CHAR(1) NULL,
    BirthDate       DATE NULL,
    HeightInches    DECIMAL(5,2) NULL,
    WeightLbs       DECIMAL(6,2) NULL,
    Active          BIT NOT NULL DEFAULT 1,
    CONSTRAINT PK_Players PRIMARY KEY (PlayerID),
    CONSTRAINT UQ_Players_NHLPlayerID UNIQUE (NHLPlayerID)
);
```

PlayerTeamHistory

```
CREATE TABLE nhl.Reference.PlayerTeamHistory
(
    PlayerTeamHistoryID BIGINT IDENTITY(1,1) NOT NULL,
    PlayerID            BIGINT NOT NULL,
    TeamID              INT NOT NULL,
    SeasonID            INT NOT NULL,
    StartDate           DATE NULL,
    EndDate             DATE NULL,
    CONSTRAINT PK_PlayerTeamHistory PRIMARY KEY (PlayerTeamHistoryID),
    CONSTRAINT FK_PTH_Player FOREIGN KEY (PlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_PTH_Team FOREIGN KEY (TeamID)
        REFERENCES nhl.Reference.Teams(TeamID),
    CONSTRAINT FK_PTH_Season FOREIGN KEY (SeasonID)
        REFERENCES nhl.Reference.Seasons(SeasonID)
);
```

Situations

```
CREATE TABLE nhl.Reference.Situations
(
```

```
    SituationID       INT IDENTITY(1,1) NOT NULL,
    SituationCode     VARCHAR(20) NOT NULL,
    Description       VARCHAR(100) NOT NULL,
    StrengthHome      TINYINT NULL,
    StrengthAway      TINYINT NULL,
    IncludeEmptyNet   BIT NOT NULL DEFAULT 1,
    CONSTRAINT PK_Situations PRIMARY KEY (SituationID),
    CONSTRAINT UQ_Situations_Code UNIQUE (SituationCode)
);
```

6. Game and Event Tables

Games

```
CREATE TABLE nhl.Game.Games
(
    GameID              BIGINT IDENTITY(1,1) NOT NULL,
    NHLGameID           INT NOT NULL,
    SeasonID            INT NOT NULL,
    GameType            VARCHAR(20) NOT NULL,
    GameDate            DATE NOT NULL,
    StartTimeUTC        DATETIME2(0) NULL,
    HomeTeamID          INT NOT NULL,
    AwayTeamID          INT NOT NULL,
    HomeScore           SMALLINT NULL,
    AwayScore           SMALLINT NULL,
    Venue               VARCHAR(200) NULL,
    GameStatus          VARCHAR(30) NULL,
    CONSTRAINT PK_Games PRIMARY KEY (GameID),
    CONSTRAINT UQ_Games_NHLGameID UNIQUE (NHLGameID),
    CONSTRAINT FK_Games_Season FOREIGN KEY (SeasonID)
        REFERENCES nhl.Reference.Seasons(SeasonID),
    CONSTRAINT FK_Games_HomeTeam FOREIGN KEY (HomeTeamID)
        REFERENCES nhl.Reference.Teams(TeamID),
    CONSTRAINT FK_Games_AwayTeam FOREIGN KEY (AwayTeamID)
        REFERENCES nhl.Reference.Teams(TeamID)
);
```

Plays

```
CREATE TABLE nhl.Game.Plays
(
    PlayID                 BIGINT IDENTITY(1,1) NOT NULL,
    GameID                 BIGINT NOT NULL,
    NHLPlayID              INT NOT NULL,
    PeriodNumber           TINYINT NOT NULL,
    PeriodTimeSeconds      SMALLINT NULL,
    PeriodTimeRemaining    SMALLINT NULL,
    EventType               VARCHAR(50) NOT NULL,
    Description            VARCHAR(1000) NULL,
    TeamID                 INT NULL,
    PlayerID               BIGINT NULL,
    SecondaryPlayerID      BIGINT NULL,
    XCoordinate            DECIMAL(6,2) NULL,
    YCoordinate            DECIMAL(6,2) NULL,
    HomeScore              SMALLINT NULL,
    AwayScore              SMALLINT NULL,
    StrengthCode           VARCHAR(20) NULL,
    CreatedAt              DATETIME2(0) NOT NULL DEFAULT SYSUTCDATETIME(),
    CONSTRAINT PK_Plays PRIMARY KEY (PlayID),
    CONSTRAINT UQ_Plays_Game_NHLPlay UNIQUE (GameID, NHLPlayID),
    CONSTRAINT FK_Plays_Game FOREIGN KEY (GameID)
        REFERENCES nhl.Game.Games(GameID),
    CONSTRAINT FK_Plays_Team FOREIGN KEY (TeamID)
        REFERENCES nhl.Reference.Teams(TeamID),
    CONSTRAINT FK_Plays_Player FOREIGN KEY (PlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_Plays_SecondaryPlayer FOREIGN KEY (SecondaryPlayerID)
        REFERENCES nhl.Reference.Players(PlayerID)
);
```

Shots

```
CREATE TABLE nhl.Game.Shots
```

```
(
    ShotID                  BIGINT IDENTITY(1,1) NOT NULL,
    PlayID                  BIGINT NOT NULL,
    GameID                  BIGINT NOT NULL,
    TeamID                  INT NOT NULL,
    ShooterPlayerID         BIGINT NULL,
    GoaliePlayerID          BIGINT NULL,
    PeriodNumber            TINYINT NOT NULL,
    PeriodTimeSeconds       SMALLINT NULL,
    ShotEventType           VARCHAR(30) NOT NULL,
    ShotType                VARCHAR(50) NULL,
    XCoordinate             DECIMAL(7,3) NULL,
    YCoordinate             DECIMAL(7,3) NULL,
    DistanceFeet             DECIMAL(7,2) NULL,
    AngleDegrees             DECIMAL(7,2) NULL,
    IsGoal                  BIT NOT NULL DEFAULT 0,
    IsBlocked               BIT NOT NULL DEFAULT 0,
    IsMissed                BIT NOT NULL DEFAULT 0,
    StrengthCode             VARCHAR(20) NULL,
    HomeScore               SMALLINT NULL,
    AwayScore               SMALLINT NULL,
    CreatedAt               DATETIME2(0) NOT NULL DEFAULT SYSUTCDATETIME(),
    CONSTRAINT PK_Shots PRIMARY KEY (ShotID),
    CONSTRAINT UQ_Shots_Play UNIQUE (PlayID),
    CONSTRAINT FK_Shots_Play FOREIGN KEY (PlayID)
        REFERENCES nhl.Game.Plays(PlayID),
    CONSTRAINT FK_Shots_Game FOREIGN KEY (GameID)
        REFERENCES nhl.Game.Games(GameID),
    CONSTRAINT FK_Shots_Team FOREIGN KEY (TeamID)
        REFERENCES nhl.Reference.Teams(TeamID),
    CONSTRAINT FK_Shots_Shooter FOREIGN KEY (ShooterPlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_Shots_Goalie FOREIGN KEY (GoaliePlayerID)
        REFERENCES nhl.Reference.Players(PlayerID)
);
```

Goals

```
CREATE TABLE nhl.Game.Goals
(
    GoalID                  BIGINT IDENTITY(1,1) NOT NULL,
    ShotID                  BIGINT NOT NULL,
    PlayID                  BIGINT NOT NULL,
    GameID                  BIGINT NOT NULL,
    TeamID                  INT NOT NULL,
    ScoringPlayerID         BIGINT NOT NULL,
    Assist1PlayerID         BIGINT NULL,
    Assist2PlayerID         BIGINT NULL,
    PeriodNumber            TINYINT NOT NULL,
    PeriodTimeSeconds       SMALLINT NULL,
    GoalType                VARCHAR(50) NULL,
    StrengthCode            VARCHAR(20) NULL,
    IsPowerPlay             BIT NOT NULL DEFAULT 0,
    IsShortHanded           BIT NOT NULL DEFAULT 0,
    IsEmptyNet              BIT NOT NULL DEFAULT 0,
    HomeScore               SMALLINT NULL,
    AwayScore               SMALLINT NULL,
    CONSTRAINT PK_Goals PRIMARY KEY (GoalID),
    CONSTRAINT UQ_Goals_Shot UNIQUE (ShotID),
    CONSTRAINT FK_Goals_Shot FOREIGN KEY (ShotID)
        REFERENCES nhl.Game.Shots(ShotID),
    CONSTRAINT FK_Goals_Play FOREIGN KEY (PlayID)
        REFERENCES nhl.Game.Plays(PlayID),
    CONSTRAINT FK_Goals_Game FOREIGN KEY (GameID)
        REFERENCES nhl.Game.Games(GameID),
    CONSTRAINT FK_Goals_Team FOREIGN KEY (TeamID)
        REFERENCES nhl.Reference.Teams(TeamID),
    CONSTRAINT FK_Goals_Scorer FOREIGN KEY (ScoringPlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_Goals_Assist1 FOREIGN KEY (Assist1PlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_Goals_Assist2 FOREIGN KEY (Assist2PlayerID)
        REFERENCES nhl.Reference.Players(PlayerID)
);
```

Shifts

```
CREATE TABLE nhl.Game.Shifts
(
    ShiftID                 BIGINT IDENTITY(1,1) NOT NULL,
    GameID                  BIGINT NOT NULL,
    PlayerID                BIGINT NOT NULL,
    TeamID                  INT NOT NULL,
    PeriodNumber            TINYINT NOT NULL,
    ShiftStartSeconds       SMALLINT NOT NULL,
    ShiftEndSeconds         SMALLINT NOT NULL,
    DurationSeconds         SMALLINT NOT NULL,
    CONSTRAINT PK_Shifts PRIMARY KEY (ShiftID),
    CONSTRAINT FK_Shifts_Game FOREIGN KEY (GameID)
        REFERENCES nhl.Game.Games(GameID),
    CONSTRAINT FK_Shifts_Player FOREIGN KEY (PlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_Shifts_Team FOREIGN KEY (TeamID)
        REFERENCES nhl.Reference.Teams(TeamID)
);
```

PlayOnIcePlayers

```
CREATE TABLE nhl.Game.PlayOnIcePlayers
(
    PlayOnIcePlayerID BIGINT IDENTITY(1,1) NOT NULL,
    PlayID            BIGINT NOT NULL,
    PlayerID          BIGINT NOT NULL,
    TeamID            INT NOT NULL,
    RoleCode           VARCHAR(20) NOT NULL,
    CONSTRAINT PK_PlayOnIcePlayers PRIMARY KEY (PlayOnIcePlayerID),
    CONSTRAINT UQ_PlayOnIcePlayers UNIQUE (PlayID, PlayerID),
    CONSTRAINT FK_POIP_Play FOREIGN KEY (PlayID)
        REFERENCES nhl.Game.Plays(PlayID),
    CONSTRAINT FK_POIP_Player FOREIGN KEY (PlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_POIP_Team FOREIGN KEY (TeamID)
        REFERENCES nhl.Reference.Teams(TeamID)
);
```

7. Official Player Statistics

PlayerGameStats

```
CREATE TABLE nhl.Stats.PlayerGameStats
(
    PlayerGameStatsID       BIGINT IDENTITY(1,1) NOT NULL,
    GameID                  BIGINT NOT NULL,
    PlayerID                BIGINT NOT NULL,
    TeamID                  INT NOT NULL,
    PositionCode            VARCHAR(5) NULL,
    Goals                   SMALLINT NULL,
    Assists                 SMALLINT NULL,
    Points                  SMALLINT NULL,
    Shots                   SMALLINT NULL,
    Hits                    SMALLINT NULL,
    Blocks                  SMALLINT NULL,
    Giveaways               SMALLINT NULL,
    Takeaways               SMALLINT NULL,
    PenaltyMinutes          SMALLINT NULL,
    FaceoffWins             SMALLINT NULL,
    FaceoffLosses           SMALLINT NULL,
    TimeOnIceSeconds        INT NULL,
    PowerPlayTOISeconds     INT NULL,
    ShortHandedTOISeconds   INT NULL,
    PowerPlayGoals          SMALLINT NULL,
    PowerPlayAssists        SMALLINT NULL,
    ShortHandedGoals        SMALLINT NULL,
    ShortHandedAssists      SMALLINT NULL,
    CreatedAt               DATETIME2(0) NOT NULL DEFAULT SYSUTCDATETIME(),
    CONSTRAINT PK_PlayerGameStats PRIMARY KEY (PlayerGameStatsID),
    CONSTRAINT UQ_PlayerGameStats UNIQUE (GameID, PlayerID, TeamID),
    CONSTRAINT FK_PGS_Game FOREIGN KEY (GameID) REFERENCES nhl.Game.Games(GameID),
    CONSTRAINT FK_PGS_Player FOREIGN KEY (PlayerID) REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_PGS_Team FOREIGN KEY (TeamID) REFERENCES nhl.Reference.Teams(TeamID)
);
```

PlayerSeasonStats

```
CREATE TABLE nhl.Stats.PlayerSeasonStats
(
    PlayerSeasonStatsID     BIGINT IDENTITY(1,1) NOT NULL,
    SeasonID                INT NOT NULL,
    PlayerID                BIGINT NOT NULL,
    TeamID                  INT NOT NULL,
    GamesPlayed             SMALLINT NULL,
    Goals                   INT NULL,
    Assists                 INT NULL,
    Points                  INT NULL,
    Shots                   INT NULL,
    Hits                    INT NULL,
    Blocks                  INT NULL,
    Giveaways               INT NULL,
    Takeaways               INT NULL,
    PenaltyMinutes          INT NULL,
    TimeOnIceSeconds        BIGINT NULL,
    PowerPlayTOISeconds     BIGINT NULL,
    ShortHandedTOISeconds   BIGINT NULL,
    PowerPlayGoals          INT NULL,
    ShortHandedGoals        INT NULL,
    CreatedAt               DATETIME2(0) NOT NULL DEFAULT SYSUTCDATETIME(),
    CONSTRAINT PK_PlayerSeasonStats PRIMARY KEY (PlayerSeasonStatsID),
    CONSTRAINT UQ_PlayerSeasonStats UNIQUE (SeasonID, PlayerID, TeamID),
    CONSTRAINT FK_PSS_Season FOREIGN KEY (SeasonID) REFERENCES nhl.Reference.Seasons(SeasonID),
    CONSTRAINT FK_PSS_Player FOREIGN KEY (PlayerID) REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_PSS_Team FOREIGN KEY (TeamID) REFERENCES nhl.Reference.Teams(TeamID)
);
```

8. Advanced Analytics Tables

CalculationVersions

```
CREATE TABLE nhl.Analytics.CalculationVersions
(
    CalculationVersionID INT IDENTITY(1,1) NOT NULL,
    MetricCode            VARCHAR(50) NOT NULL,
    VersionName           VARCHAR(100) NOT NULL,
    ModelDescription      VARCHAR(MAX) NULL,
    CreatedAt              DATETIME2(0) NOT NULL DEFAULT SYSUTCDATETIME(),
    IsActive              BIT NOT NULL DEFAULT 1,
    CONSTRAINT PK_CalculationVersions PRIMARY KEY (CalculationVersionID),
    CONSTRAINT UQ_CalculationVersions UNIQUE (MetricCode, VersionName)
);
```

ShotExpectedGoals

```
CREATE TABLE nhl.Analytics.ShotExpectedGoals
(
    ShotExpectedGoalsID   BIGINT IDENTITY(1,1) NOT NULL,
    ShotID                BIGINT NOT NULL,
    CalculationVersionID  INT NOT NULL,
    ExpectedGoals         DECIMAL(12,8) NOT NULL,
    CalculatedAt          DATETIME2(0) NOT NULL DEFAULT SYSUTCDATETIME(),
    CONSTRAINT PK_ShotExpectedGoals PRIMARY KEY (ShotExpectedGoalsID),
    CONSTRAINT UQ_ShotExpectedGoals UNIQUE (ShotID, CalculationVersionID),
    CONSTRAINT FK_ShotXG_Shot FOREIGN KEY (ShotID)
        REFERENCES nhl.Game.Shots(ShotID),
    CONSTRAINT FK_ShotXG_Version FOREIGN KEY (CalculationVersionID)
        REFERENCES nhl.Analytics.CalculationVersions(CalculationVersionID)
);
```

PlayerGameAdvancedStats

```
CREATE TABLE nhl.Analytics.PlayerGameAdvancedStats
(
    PlayerGameAdvancedStatsID BIGINT IDENTITY(1,1) NOT NULL,
    GameID                   BIGINT NOT NULL,
    PlayerID                 BIGINT NOT NULL,
    TeamID                   INT NOT NULL,
    SituationID              INT NOT NULL,
    CalculationVersionID     INT NOT NULL,

    IndividualCorsiFor       INT NULL,
    IndividualFenwickFor     INT NULL,
    IndividualShots          INT NULL,
    IndividualGoals          INT NULL,
    IndividualExpectedGoals  DECIMAL(12,4) NULL,
    IndividualScoringChances INT NULL,
    IndividualHighDangerChances INT NULL,

    IndividualCorsiForPct    DECIMAL(7,4) NULL,
    ShootingPercentage       DECIMAL(7,4) NULL,

    CalculatedAt             DATETIME2(0) NOT NULL DEFAULT SYSUTCDATETIME(),

    CONSTRAINT PK_PlayerGameAdvancedStats PRIMARY KEY (PlayerGameAdvancedStatsID),
    CONSTRAINT UQ_PlayerGameAdvancedStats
        UNIQUE (GameID, PlayerID, TeamID, SituationID, CalculationVersionID),

    CONSTRAINT FK_PGAS_Game FOREIGN KEY (GameID)
        REFERENCES nhl.Game.Games(GameID),
    CONSTRAINT FK_PGAS_Player FOREIGN KEY (PlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_PGAS_Team FOREIGN KEY (TeamID)
        REFERENCES nhl.Reference.Teams(TeamID),
    CONSTRAINT FK_PGAS_Situation FOREIGN KEY (SituationID)
        REFERENCES nhl.Reference.Situations(SituationID),
    CONSTRAINT FK_PGAS_Version FOREIGN KEY (CalculationVersionID)
        REFERENCES nhl.Analytics.CalculationVersions(CalculationVersionID)
);
```

PlayerGameOnIceStats

```
CREATE TABLE nhl.Analytics.PlayerGameOnIceStats
(
    PlayerGameOnIceStatsID BIGINT IDENTITY(1,1) NOT NULL,
    GameID                 BIGINT NOT NULL,
    PlayerID               BIGINT NOT NULL,
```

```
    TeamID                 INT NOT NULL,
    SituationID            INT NOT NULL,
    CalculationVersionID   INT NOT NULL,

    TimeOnIceSeconds       INT NULL,

    CorsiFor                INT NULL,
    CorsiAgainst           INT NULL,
    CorsiForPct             DECIMAL(7,4) NULL,

    FenwickFor              INT NULL,
    FenwickAgainst         INT NULL,
    FenwickForPct           DECIMAL(7,4) NULL,

    ShotsFor                INT NULL,
    ShotsAgainst            INT NULL,

    GoalsFor                INT NULL,
    GoalsAgainst            INT NULL,

    ExpectedGoalsFor       DECIMAL(12,4) NULL,
    ExpectedGoalsAgainst   DECIMAL(12,4) NULL,
    ExpectedGoalsPct        DECIMAL(7,4) NULL,

    ScoringChancesFor      INT NULL,
    ScoringChancesAgainst  INT NULL,
    HighDangerChancesFor   INT NULL,
    HighDangerChancesAgainst INT NULL,

    PDO                   DECIMAL(8,4) NULL,
    OnIceShootingPct      DECIMAL(8,4) NULL,
    OnIceSavePct          DECIMAL(8,4) NULL,

    CalculatedAt           DATETIME2(0) NOT NULL DEFAULT SYSUTCDATETIME(),

    CONSTRAINT PK_PlayerGameOnIceStats PRIMARY KEY (PlayerGameOnIceStatsID),
    CONSTRAINT UQ_PlayerGameOnIceStats
        UNIQUE (GameID, PlayerID, TeamID, SituationID, CalculationVersionID),

    CONSTRAINT FK_PGOIS_Game FOREIGN KEY (GameID)
        REFERENCES nhl.Game.Games(GameID),
    CONSTRAINT FK_PGOIS_Player FOREIGN KEY (PlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_PGOIS_Team FOREIGN KEY (TeamID)
        REFERENCES nhl.Reference.Teams(TeamID),
    CONSTRAINT FK_PGOIS_Situation FOREIGN KEY (SituationID)
        REFERENCES nhl.Reference.Situations(SituationID),
    CONSTRAINT FK_PGOIS_Version FOREIGN KEY (CalculationVersionID)
        REFERENCES nhl.Analytics.CalculationVersions(CalculationVersionID)
);
```

GoalieGameAdvancedStats

```
CREATE TABLE nhl.Analytics.GoalieGameAdvancedStats
(
    GoalieGameAdvancedStatsID BIGINT IDENTITY(1,1) NOT NULL,
    GameID                   BIGINT NOT NULL,
    GoaliePlayerID           BIGINT NOT NULL,
    TeamID                   INT NOT NULL,
    SituationID              INT NOT NULL,
    CalculationVersionID     INT NOT NULL,

    ShotsAgainst             INT NULL,
    Saves                    INT NULL,
    GoalsAgainst             INT NULL,
    SavePercentage           DECIMAL(8,5) NULL,

    ExpectedGoalsAgainst     DECIMAL(12,4) NULL,
    GoalsSavedAboveExpected  DECIMAL(12,4) NULL,
    ExpectedSavePercentage   DECIMAL(8,5) NULL,

    HighDangerShotsAgainst   INT NULL,
    HighDangerSaves          INT NULL,
    HighDangerGoalsAgainst   INT NULL,

    ReboundShotsAgainst      INT NULL,
    ReboundGoalsAgainst      INT NULL,
    RushShotsAgainst         INT NULL,
```

```
    RushGoalsAgainst         INT NULL,

    CalculatedAt             DATETIME2(0) NOT NULL DEFAULT SYSUTCDATETIME(),

    CONSTRAINT PK_GoalieGameAdvancedStats PRIMARY KEY (GoalieGameAdvancedStatsID),
    CONSTRAINT UQ_GoalieGameAdvancedStats
        UNIQUE (GameID, GoaliePlayerID, TeamID, SituationID, CalculationVersionID),

    CONSTRAINT FK_GGAS_Game FOREIGN KEY (GameID)
        REFERENCES nhl.Game.Games(GameID),
    CONSTRAINT FK_GGAS_Goalie FOREIGN KEY (GoaliePlayerID)
        REFERENCES nhl.Reference.Players(PlayerID),
    CONSTRAINT FK_GGAS_Team FOREIGN KEY (TeamID)
        REFERENCES nhl.Reference.Teams(TeamID),
    CONSTRAINT FK_GGAS_Situation FOREIGN KEY (SituationID)
        REFERENCES nhl.Reference.Situations(SituationID),
    CONSTRAINT FK_GGAS_Version FOREIGN KEY (CalculationVersionID)
        REFERENCES nhl.Analytics.CalculationVersions(CalculationVersionID)
);
```

9. What Should Be Stored vs. Calculated?

Data / Metric	Store?	Reason
Raw NHL API response	YES	Allows reprocessing when parsers/calculations change.
Play-by-play event	YES	Fundamental source data.
Shot event	YES	Foundation for Corsi, Fenwick, xG and goalie analytics.
Shift	YES	Required to reconstruct time-on-ice and on-ice state.
Players on ice	YES	Required for on-ice player attribution.
Official NHL player stats	YES	Preserve official values for comparison.
Shot-level xG	YES	Expensive/model-based and useful for downstream aggregation.
Player-game Corsi	YES	Fast querying; reproducible from events.
Player-game on-ice xG	YES	Fast querying; expensive to reconstruct repeatedly.
Player-season Corsi	CALCULATE/MATERIALIZE	Can aggregate from game-level data; optionally store for performance.
Career totals	CALCULATE	Should generally be aggregation/query logic.
Custom date-range metrics	CALCULATE	Keep event-level data to support arbitrary analysis.

10. Calculation Model Strategy

• **Individual statistics:** identify the shooter/player responsible for an event and aggregate that player's actions.

• **On-ice statistics:** identify every player on the ice at the event, then attribute the event to each appropriate player.

• **Goalie statistics:** identify the defending goalie for each shot and combine shot outcome with shot-level xG.

• **Corsi:** define the exact event types included in the calculation and calculate For/Against consistently by situation.

• **Fenwick:** use unblocked shot attempts according to the selected calculation definition.

• **xG:** calculate a probability for every relevant shot and retain the model version.

• **GSAx:** derive from expected goals against versus actual goals against for the goalie.

• **PDO:** calculate from on-ice shooting percentage plus on-ice save percentage according to the selected definition.

• **Situation filtering:** every advanced-stat calculation should be capable of filtering by 5v5, power play, penalty kill, 4v4, 3v3, empty-net, score state, and other supported situations.

11. Critical Requirement: Event Attribution

The most important part of the database is not the advanced-stat tables themselves. It is the ability to accurately attribute every event to the players and teams involved. For every shot, the analytics engine should be able to determine the shooting team, shooter, defending team, goalie, game situation, score state, coordinates, and players on the ice. This allows the same event data to generate many different analytical views without duplicating the underlying facts.

12. Indexing Requirements

```
CREATE INDEX IX_Plays_GameID
ON nhl.Game.Plays(GameID);

CREATE INDEX IX_Shots_GameID
```

```
ON nhl.Game.Shots(GameID);

CREATE INDEX IX_Shots_TeamID
ON nhl.Game.Shots(TeamID);

CREATE INDEX IX_Shots_Shooter
ON nhl.Game.Shots(ShooterPlayerID);

CREATE INDEX IX_Shots_Goalie
ON nhl.Game.Shots(GoaliePlayerID);

CREATE INDEX IX_Goals_ScoringPlayer
ON nhl.Game.Goals(ScoringPlayerID);

CREATE INDEX IX_Shifts_PlayerGame
ON nhl.Game.Shifts(PlayerID, GameID);

CREATE INDEX IX_PlayOnIcePlayers_Play
ON nhl.Game.PlayOnIcePlayers(PlayID);

CREATE INDEX IX_PlayOnIcePlayers_Player
ON nhl.Game.PlayOnIcePlayers(PlayerID);

CREATE INDEX IX_PlayerGameStats_Player
ON nhl.Stats.PlayerGameStats(PlayerID);

CREATE INDEX IX_PlayerSeasonStats_Player
ON nhl.Stats.PlayerSeasonStats(PlayerID);

CREATE INDEX IX_PlayerGameAdvanced_Player
ON nhl.Analytics.PlayerGameAdvancedStats(PlayerID);

CREATE INDEX IX_PlayerGameOnIce_Player
ON nhl.Analytics.PlayerGameOnIceStats(PlayerID);

CREATE INDEX IX_GoalieGameAdvanced_Goalie
ON nhl.Analytics.GoalieGameAdvancedStats(GoaliePlayerID);
```

13. Ingestion Workflow

```
1. Discover/import season and game information.
2. Create or update Teams, Players and Games.
3. Retrieve and archive raw NHL API responses.
4. Parse play-by-play into Plays.
5. Parse shot events into Shots.
6. Convert goal events into Goals.
7. Parse shifts.
8. Determine players on ice for each relevant event.
9. Import official PlayerGameStats.
10. Import/update PlayerSeasonStats.
11. Calculate shot characteristics needed by analytics.
12. Run the configured xG model against eligible shots.
13. Calculate individual player game analytics.
14. Calculate player on-ice game analytics.
15. Calculate goalie game analytics.
16. Store the results with the appropriate CalculationVersionID.
17. Validate totals against official NHL statistics where applicable.
18. Mark ingestion/recalculation status SUCCESS or record an error for retry.
```

14. Idempotency and Recalculation

• NHLGameID uniquely identifies a game.

• (GameID, NHLPlayID) uniquely identifies a play.

• PlayID uniquely identifies the normalized shot associated with that play.

• (GameID, PlayerID, TeamID) uniquely identifies a player-game official-stat record.

• (SeasonID, PlayerID, TeamID) uniquely identifies a player-season official-stat record.

• (ShotID, CalculationVersionID) uniquely identifies an xG calculation.

• (GameID, PlayerID, TeamID, SituationID, CalculationVersionID) uniquely identifies individual/on-ice player-game analytics.

• (GameID, GoaliePlayerID, TeamID, SituationID, CalculationVersionID) uniquely identifies goalie-game analytics.

• Changing an xG model should create a new CalculationVersion rather than overwrite previous model results.

• Changing a Corsi/Fenwick definition should likewise be represented by a calculation version if the resulting values are not directly comparable.

• The system should support recalculating advanced statistics without re-downloading the NHL API data.

15. Claude Code Implementation Instructions

• Treat this document as the target database specification.

• Create the database schemas if they do not already exist.

• Create tables in dependency order: Reference → Game → Stats → Analytics → Ingestion.

• Use SQL Server-compatible T-SQL only.

• Create all primary keys, unique constraints, foreign keys and indexes specified above.

• Do not use player names or team abbreviations as relational keys.

• Keep NHL external IDs in dedicated columns.

• Use DATETIME2 and SYSUTCDATETIME() for timestamps.

• Make ingestion safe to run repeatedly.

• Create seed data for common Situations and initial CalculationVersions.

• Do not hard-code advanced statistics into the ingestion parser; implement an analytics/calculation layer that reads normalized event data.

• Keep raw event data independent from derived metrics so analytical definitions can change later.

• Allow multiple calculation versions to coexist.

• Do not add video/media tables to this database design.

16. Recommended V1 Implementation

Start with one complete NHL game and make the entire pipeline work end-to-end. V1 should ingest teams, players, game metadata, play-by-play, shots, goals, shifts, players-on-ice, official player statistics, and raw API responses. Then calculate individual player statistics, on-ice statistics, and goalie statistics for that one game. Validate the basic totals before scaling to a full season. Once the event attribution is reliable, add season-level materializations and more sophisticated models.

17. Future Analytics Extensions

• Player-season and team-season advanced statistics.

• On-ice Corsi/xG by teammate and opponent.

• Line-combination analytics.

• Score-state and zone-based splits.

• Rush, rebound, and pre-shot movement models.

• High-danger and slot-based models.

• Goals above expected for skaters.

• Goals saved above expected for goalies.

• RAPM/GAR/WAR-style models.

• Rolling 5/10/20-game statistics.

• Career analytics.

• Custom user-defined metrics and model versions.